# determinante

January 13, 2026

[1]:
```
%load_ext autoreload
```

## 0.1 Nombre: Luis Lema

## 0.2 Fecha: 13/01/2025

## 0.3 Repositorio: https://github.com/LuisALema/Metodos_Numericos_2025B/tree/main/

## 0.4 Determinante

[7]:
```
# ============================================================
# CELDA 1 (una sola vez): IMPORTS + FUNCIONES PRINCIPALES
# ============================================================
%autoreload 2
import numpy as np
from src import (
    descomposicion_LU,
    resolver_LU,
)

def inversa_por_LU(A: list[list[float]]) -> np.ndarray:
    A_np = np.array(A, dtype=float)
    n = A_np.shape[0]
    assert A_np.shape[0] == A_np.shape[1], "A debe ser cuadrada."

    L, U = descomposicion_LU(A_np)

    invA = np.zeros((n, n), dtype=float)
    for i in range(n):
        e = np.zeros((n, 1), dtype=float)
        e[i, 0] = 1.0
        x = resolver_LU(L, U, e)
        invA[:, i] = x.flatten()

    return invA


def procesar_matriz(A: list[list[float]], b_list: list[list[float]], nombre:␣
 ↪str):
```

```
    A_np = np.array(A, dtype=float)
    n = A_np.shape[0]

    print(f"\n==================== {nombre} ====================")

    # Inversa
    invA = inversa_por_LU(A)
    print(f"\nInversa de {nombre}:\n{invA}")

    # LU
    L, U = descomposicion_LU(A_np)
    print(f"\nL de {nombre}:\n{L}")
    print(f"\nU de {nombre}:\n{U}")

    # Soluciones Ax=b
    for i, b in enumerate(b_list, start=1):
        b_np = np.array(b, dtype=float).reshape(-1, 1)

        if b_np.shape[0] != n:
            print(f"\n(b{i}) NO se puede: b tiene {b_np.shape[0]} filas y␣
↪{nombre} es {n}x{n}")
            continue

        x = resolver_LU(L, U, b_np)
        print(f"\nSolución para b{i} (de {nombre}):\n{b_np}\n=> x =\n{x}")
```

# 1 Ejercicio 1

[ ]:
```
A1 = [
    [1, 3, 4],
    [2, 1, 3],
    [4, 2, 1],
]

b1_3 = [1, 2, 4]
b2_3 = [3, -5, 2]
b3_3 = [7, 8, -1]

procesar_matriz(A1, [b1_3, b2_3, b3_3], "A1")
```

```
==================== A1 ====================
[01-13 18:28:47][INFO]
[[  1.   3.   4.]
 [  0.  -5.  -5.]
 [  0. -10. -15.]]
[01-13 18:28:47][INFO]
```

```
[[ 1.  3.  4.]
 [ 0. -5. -5.]
 [ 0.  0. -5.]]
[01-13 18:28:47][INFO]
[[ 1.  3.  4.]
 [ 0. -5. -5.]
 [ 0.  0. -5.]]
[01-13 18:28:47][INFO] Sustitución hacia adelante
[01-13 18:28:47][INFO] y =
[[ 1.]
 [-2.]
 [ 0.]]
[01-13 18:28:47][INFO] Sustitución hacia atrás
[01-13 18:28:47][INFO] i = 1
[01-13 18:28:47][INFO] suma = [0.]
[01-13 18:28:47][INFO] U[i, i] = -5.0

[01-13 18:28:47][INFO] y[i] = [-2.]
[01-13 18:28:47][INFO] i = 0
[01-13 18:28:47][INFO] i = 0
[01-13 18:28:47][INFO] suma = [1.2]
[01-13 18:28:47][INFO] U[i, i] = 1.0
[01-13 18:28:47][INFO] y[i] = [1.]
[01-13 18:28:48][INFO] Sustitución hacia adelante
[01-13 18:28:48][INFO] y =
[[ 0.]
 [ 1.]
 [-2.]]
[01-13 18:28:48][INFO] Sustitución hacia atrás
[01-13 18:28:48][INFO] i = 1
[01-13 18:28:48][INFO] suma = [-2.]
[01-13 18:28:48][INFO] U[i, i] = -5.0
[01-13 18:28:48][INFO] y[i] = [1.]
[01-13 18:28:48][INFO] i = 0
[01-13 18:28:48][INFO] suma = [-0.2]
[01-13 18:28:48][INFO] U[i, i] = 1.0
[01-13 18:28:48][INFO] y[i] = [0.]
[01-13 18:28:48][INFO] Sustitución hacia adelante
[01-13 18:28:48][INFO] y =
[[0.]
 [0.]
 [1.]]
[01-13 18:28:48][INFO] Sustitución hacia atrás
[01-13 18:28:48][INFO] i = 1
[01-13 18:28:48][INFO] suma = [1.]
[01-13 18:28:48][INFO] U[i, i] = -5.0
[01-13 18:28:48][INFO] y[i] = [0.]
[01-13 18:28:48][INFO] i = 0
```

```
[01-13 18:28:48][INFO] suma = [-0.2]
[01-13 18:28:48][INFO] U[i, i] = 1.0
[01-13 18:28:48][INFO] y[i] = [0.]

Inversa de A1:
[[-0.2  0.2  0.2]
 [ 0.4 -0.6  0.2]
 [-0.   0.4 -0.2]]
[01-13 18:28:48][INFO]
[[  1.   3.   4.]
 [  0.  -5.  -5.]
 [  0. -10. -15.]]
[01-13 18:28:48][INFO]
[[ 1.  3.  4.]
 [ 0. -5. -5.]
 [ 0.  0. -5.]]
[01-13 18:28:48][INFO]
[[ 1.  3.  4.]
 [ 0. -5. -5.]
 [ 0.  0. -5.]]

L de A1:
[[1. 0. 0.]
 [2. 1. 0.]
 [4. 2. 1.]]

U de A1:
[[ 1.  3.  4.]
 [ 0. -5. -5.]
 [ 0.  0. -5.]]
[01-13 18:28:48][INFO] Sustitución hacia adelante
[01-13 18:28:48][INFO] y =
[[1.]
 [0.]
 [0.]]
[01-13 18:28:48][INFO] Sustitución hacia atrás
[01-13 18:28:48][INFO] i = 1
[01-13 18:28:48][INFO] suma = [0.]
[01-13 18:28:48][INFO] U[i, i] = -5.0
[01-13 18:28:48][INFO] y[i] = [0.]
[01-13 18:28:48][INFO] i = 0
[01-13 18:28:48][INFO] suma = [0.]
[01-13 18:28:48][INFO] U[i, i] = 1.0
[01-13 18:28:48][INFO] y[i] = [1.]

Solución para b1 (de A1):
[[1.]
 [2.]
```

```
 [4.]]
=> x =
[[ 1.]
 [-0.]
 [-0.]]
[01-13 18:28:48][INFO] Sustitución hacia adelante
[01-13 18:28:48][INFO] y =
[[  3.]
 [-11.]
 [ 12.]]
[01-13 18:28:48][INFO] Sustitución hacia atrás
[01-13 18:28:48][INFO] i = 1
[01-13 18:28:48][INFO] suma = [12.]
[01-13 18:28:48][INFO] U[i, i] = -5.0
[01-13 18:28:48][INFO] y[i] = [-11.]
[01-13 18:28:48][INFO] i = 0
[01-13 18:28:48][INFO] suma = [4.2]
[01-13 18:28:48][INFO] U[i, i] = 1.0
[01-13 18:28:48][INFO] y[i] = [3.]

Solución para b2 (de A1):
[[ 3.]
 [-5.]
 [ 2.]]
=> x =
[[-1.2]
 [ 4.6]
 [-2.4]]
[01-13 18:28:48][INFO] Sustitución hacia adelante
[01-13 18:28:48][INFO] y =
[[  7.]
 [ -6.]
 [-17.]]
[01-13 18:28:48][INFO] Sustitución hacia atrás
[01-13 18:28:48][INFO] i = 1
[01-13 18:28:48][INFO] suma = [-17.]
[01-13 18:28:48][INFO] U[i, i] = -5.0
[01-13 18:28:48][INFO] y[i] = [-6.]
[01-13 18:28:48][INFO] i = 0
[01-13 18:28:48][INFO] suma = [7.]
[01-13 18:28:48][INFO] U[i, i] = 1.0
[01-13 18:28:48][INFO] y[i] = [7.]

Solución para b3 (de A1):
[[ 7.]
 [ 8.]
 [-1.]]
=> x =
```

```
[[ 8.8817842e-16]
 [-2.2000000e+00]
 [ 3.4000000e+00]]
```

## 2 Ejercicio 2

[9]:
```
A2 = [
    [1, 2, 3],
    [0, 1, 4],
    [5, 6, 0],
]

b1_3 = [1, 2, 4]
b2_3 = [3, -5, 2]
b3_3 = [7, 8, -1]

procesar_matriz(A2, [b1_3, b2_3, b3_3], "A2")
```

```
==================== A2 ====================
[01-13 18:29:20][INFO]
[[  1.   2.   3.]
 [  0.   1.   4.]
 [  0.  -4. -15.]]
[01-13 18:29:20][INFO]
[[1. 2. 3.]
 [0. 1. 4.]
 [0. 0. 1.]]
[01-13 18:29:20][INFO]
[[1. 2. 3.]
 [0. 1. 4.]
 [0. 0. 1.]]
[01-13 18:29:20][INFO] Sustitución hacia adelante
[01-13 18:29:20][INFO] y =
[[ 1.]
 [ 0.]
 [-5.]]
[01-13 18:29:20][INFO] Sustitución hacia atrás
[01-13 18:29:20][INFO] i = 1
[01-13 18:29:20][INFO] suma = [-20.]
[01-13 18:29:20][INFO] U[i, i] = 1.0
[01-13 18:29:20][INFO] y[i] = [0.]
[01-13 18:29:20][INFO] i = 0

[01-13 18:29:20][INFO] suma = [25.]
[01-13 18:29:20][INFO] U[i, i] = 1.0
[01-13 18:29:20][INFO] y[i] = [1.]
[01-13 18:29:20][INFO] U[i, i] = 1.0
```

```
[01-13 18:29:20][INFO] y[i] = [1.]
[01-13 18:29:20][INFO] Sustitución hacia adelante
[01-13 18:29:20][INFO] y =
[[0.]
 [1.]
 [4.]]
[01-13 18:29:20][INFO] Sustitución hacia atrás
[01-13 18:29:20][INFO] i = 1
[01-13 18:29:20][INFO] suma = [16.]
[01-13 18:29:20][INFO] U[i, i] = 1.0
[01-13 18:29:20][INFO] y[i] = [1.]
[01-13 18:29:20][INFO] i = 0
[01-13 18:29:20][INFO] suma = [-18.]
[01-13 18:29:20][INFO] U[i, i] = 1.0
[01-13 18:29:20][INFO] y[i] = [0.]
[01-13 18:29:20][INFO] Sustitución hacia adelante
[01-13 18:29:20][INFO] y =
[[0.]
 [0.]
 [1.]]
[01-13 18:29:20][INFO] Sustitución hacia atrás
[01-13 18:29:20][INFO] i = 1
[01-13 18:29:20][INFO] suma = [4.]
[01-13 18:29:20][INFO] U[i, i] = 1.0
[01-13 18:29:20][INFO] y[i] = [0.]
[01-13 18:29:20][INFO] i = 0
[01-13 18:29:20][INFO] suma = [-5.]
[01-13 18:29:20][INFO] U[i, i] = 1.0
[01-13 18:29:20][INFO] y[i] = [0.]

Inversa de A2:
[[-24.  18.   5.]
 [ 20. -15.  -4.]
 [ -5.   4.   1.]]
[01-13 18:29:20][INFO]
[[  1.   2.   3.]
 [  0.   1.   4.]
 [  0.  -4. -15.]]
[01-13 18:29:20][INFO]
[[1. 2. 3.]
 [0. 1. 4.]
 [0. 0. 1.]]
[01-13 18:29:20][INFO]
[[1. 2. 3.]
 [0. 1. 4.]
 [0. 0. 1.]]

L de A2:
```

```
[[ 1.  0.  0.]
 [ 0.  1.  0.]
 [ 5. -4.  1.]]

U de A2:
[[1. 2. 3.]
 [0. 1. 4.]
 [0. 0. 1.]]
[01-13 18:29:20][INFO] Sustitución hacia adelante
[01-13 18:29:20][INFO] y =
[[1.]
 [2.]
 [7.]]
[01-13 18:29:20][INFO] Sustitución hacia atrás
[01-13 18:29:20][INFO] i = 1
[01-13 18:29:20][INFO] suma = [28.]
[01-13 18:29:20][INFO] U[i, i] = 1.0
[01-13 18:29:20][INFO] y[i] = [2.]
[01-13 18:29:20][INFO] i = 0
[01-13 18:29:20][INFO] suma = [-31.]
[01-13 18:29:20][INFO] U[i, i] = 1.0
[01-13 18:29:20][INFO] y[i] = [1.]

Solución para b1 (de A2):
[[1.]
 [2.]
 [4.]]
=> x =
[[ 32.]
 [-26.]
 [  7.]]
[01-13 18:29:20][INFO] Sustitución hacia adelante
[01-13 18:29:20][INFO] y =
[[  3.]
 [ -5.]
 [-33.]]
[01-13 18:29:20][INFO] Sustitución hacia atrás
[01-13 18:29:20][INFO] i = 1
[01-13 18:29:20][INFO] suma = [-132.]
[01-13 18:29:20][INFO] U[i, i] = 1.0
[01-13 18:29:20][INFO] y[i] = [-5.]
[01-13 18:29:20][INFO] i = 0
[01-13 18:29:20][INFO] suma = [155.]
[01-13 18:29:20][INFO] U[i, i] = 1.0
[01-13 18:29:20][INFO] y[i] = [3.]

Solución para b2 (de A2):
[[ 3.]
```

```
 [-5.]
 [ 2.]]
=> x =
[[-152.]
 [ 127.]
 [ -33.]]
[01-13 18:29:20][INFO] Sustitución hacia adelante
[01-13 18:29:20][INFO] y =
[[ 7.]
 [ 8.]
 [-4.]]
[01-13 18:29:20][INFO] Sustitución hacia atrás
[01-13 18:29:20][INFO] i = 1
[01-13 18:29:20][INFO] suma = [-16.]
[01-13 18:29:20][INFO] U[i, i] = 1.0
[01-13 18:29:20][INFO] y[i] = [8.]
[01-13 18:29:20][INFO] i = 0
[01-13 18:29:20][INFO] suma = [36.]
[01-13 18:29:20][INFO] U[i, i] = 1.0
[01-13 18:29:20][INFO] y[i] = [7.]

Solución para b3 (de A2):
[[ 7.]
 [ 8.]
 [-1.]]
=> x =
[[-29.]
 [ 24.]
 [ -4.]]
```

[10]:
```
A3 = [
    [4, 2, 1],
    [2, 1, 3],
    [1, 3, 4],
]

b1_3 = [1, 2, 4]
b2_3 = [3, -5, 2]
b3_3 = [7, 8, -1]

procesar_matriz(A3, [b1_3, b2_3, b3_3], "A3")
```

```
==================== A3 ====================
[01-13 18:29:39][INFO]
[[4.   2.   1.  ]
 [0.   0.   2.5 ]
 [0.   2.5  3.75]]
```

```
---------------------------------------------------------------------------
ValueError                                Traceback (most recent call last)
Cell In[10], line 11
      8 b2_3 = [3, -5, 2]
      9 b3_3 = [7, 8, -1]
---> 11 procesar_matriz(A3, [b1_3, b2_3, b3_3],      )

Cell In[7], line 35, in procesar_matriz(A, b_list, nombre)
     32 print(f"\n=================== {nombre} ====================")
     34 # Inversa
---> 35 invA = inversa_por_LU(A)
     36 print(f"\nInversa de {nombre}:\n{invA}")
     38 # LU

Cell In[7], line 16, in inversa_por_LU(A)
     13 n = A_np.shape[0]
     14 assert A_np.shape[0] == A_np.shape[1], "A debe ser cuadrada."
---> 16 L, U = descomposicion_LU(A_np)
     18 invA = np.zeros((n, n), dtype=float)
     19 for i in range(n):

File c:\Users\bzbzb\Documents\GitHub\Metodos_Numericos_2025B\Talleres\Taller␣
↪2B\src\linear_syst_methods.py:127, in descomposicion_LU(A)
    123 for i in range(0, n):  # loop por columna
    124
    125     # --- deterimnar pivote
    126     if A[i, i] == 0:
--> 127         raise ValueError("No existe solución única.")
    129     # --- Eliminación: loop por fila
    130     L[i, i] = 1

ValueError: No existe solución única.
```

[11]:
```
A4 = [
    [2, 4, 6, 1],
    [4, 7, 5, -6],
    [2, 5, 18, 10],
    [6, 12, 38, 16],
]

b1_4 = [1, 2, 4, 5]
b2_4 = [3, -5, 2, 6]
b3_4 = [7, 8, -1, 0]

procesar_matriz(A4, [b1_4, b2_4, b3_4], "A4")
```

```
==================== A4 ====================
[01-13 18:30:06][INFO]
[[ 2.  4.  6.  1.]
 [ 0. -1. -7. -8.]
 [ 0.  1. 12.  9.]
 [ 0.  0. 20. 13.]]
[01-13 18:30:06][INFO]
[[ 2.  4.  6.  1.]
 [ 0. -1. -7. -8.]
 [ 0.  0.  5.  1.]
 [ 0.  0. 20. 13.]]
[01-13 18:30:06][INFO]
[[ 2.  4.  6.  1.]
 [ 0. -1. -7. -8.]
 [ 0.  0.  5.  1.]
 [ 0.  0.  0.  9.]]
[01-13 18:30:06][INFO]
[[ 2.  4.  6.  1.]
 [ 0. -1. -7. -8.]
 [ 0.  0.  5.  1.]
 [ 0.  0.  0.  9.]]
[01-13 18:30:06][INFO] Sustitución hacia adelante
[01-13 18:30:06][INFO] y =
[[ 1.]
 [-2.]
 [-3.]
 [ 9.]]
[01-13 18:30:06][INFO] Sustitución hacia atrás
[01-13 18:30:06][INFO] i = 2
[01-13 18:30:06][INFO] suma = [1.]
[01-13 18:30:06][INFO] U[i, i] = 5.0
[01-13 18:30:06][INFO] y[i] = [-3.]
[01-13 18:30:06][INFO] i = 1
[01-13 18:30:06][INFO] suma = [-2.4]
[01-13 18:30:06][INFO] U[i, i] = -1.0
[01-13 18:30:06][INFO] y[i] = [-2.]
[01-13 18:30:06][INFO] i = 0
[01-13 18:30:06][INFO] suma = [-5.4]
[01-13 18:30:06][INFO] U[i, i] = 2.0
[01-13 18:30:06][INFO] y[i] = [1.]
[01-13 18:30:06][INFO] Sustitución hacia adelante
[01-13 18:30:06][INFO] y =
[[ 0.]
 [ 1.]
 [ 1.]
 [-4.]]
[01-13 18:30:06][INFO] Sustitución hacia atrás
```

```
[01-13 18:30:06][INFO] i = 2
[01-13 18:30:06][INFO] suma = [-0.44444444]
[01-13 18:30:06][INFO] U[i, i] = 5.0
[01-13 18:30:06][INFO] y[i] = [1.]
[01-13 18:30:06][INFO] i = 1
[01-13 18:30:06][INFO] suma = [1.53333333]
[01-13 18:30:06][INFO] U[i, i] = -1.0
[01-13 18:30:06][INFO] y[i] = [1.]
[01-13 18:30:06][INFO] i = 0
[01-13 18:30:06][INFO] suma = [3.42222222]
[01-13 18:30:06][INFO] U[i, i] = 2.0
[01-13 18:30:06][INFO] y[i] = [0.]
[01-13 18:30:06][INFO] Sustitución hacia adelante
[01-13 18:30:06][INFO] y =
[[ 0.]
 [ 0.]
 [ 1.]
 [-4.]]
[01-13 18:30:07][INFO] Sustitución hacia atrás
[01-13 18:30:07][INFO] i = 2
[01-13 18:30:07][INFO] suma = [-0.44444444]
[01-13 18:30:07][INFO] U[i, i] = 5.0
[01-13 18:30:07][INFO] y[i] = [1.]
[01-13 18:30:07][INFO] i = 1
[01-13 18:30:07][INFO] suma = [1.53333333]
[01-13 18:30:07][INFO] U[i, i] = -1.0
[01-13 18:30:07][INFO] y[i] = [0.]
[01-13 18:30:07][INFO] i = 0
[01-13 18:30:07][INFO] suma = [7.42222222]
[01-13 18:30:07][INFO] U[i, i] = 2.0
[01-13 18:30:07][INFO] y[i] = [0.]
[01-13 18:30:07][INFO] Sustitución hacia adelante
[01-13 18:30:07][INFO] y =
[[0.]
 [0.]
 [0.]
 [1.]]
[01-13 18:30:07][INFO] Sustitución hacia atrás
[01-13 18:30:07][INFO] i = 2
[01-13 18:30:07][INFO] suma = [0.11111111]
[01-13 18:30:07][INFO] U[i, i] = 5.0
[01-13 18:30:07][INFO] y[i] = [0.]
[01-13 18:30:07][INFO] i = 1
[01-13 18:30:07][INFO] suma = [-0.73333333]
[01-13 18:30:07][INFO] U[i, i] = -1.0
[01-13 18:30:07][INFO] y[i] = [0.]
[01-13 18:30:07][INFO] i = 0
[01-13 18:30:07][INFO] suma = [-2.95555556]
```

```
[01-13 18:30:07][INFO] U[i, i] = 2.0
[01-13 18:30:07][INFO] y[i] = [0.]

Inversa de A4:
[[ 3.2        -1.71111111 -3.71111111  1.47777778]
 [-0.4         0.53333333  1.53333333 -0.73333333]
 [-0.8         0.28888889  0.28888889 -0.02222222]
 [ 1.         -0.44444444 -0.44444444  0.11111111]]
[01-13 18:30:07][INFO]
[[ 2.  4.  6.  1.]
 [ 0. -1. -7. -8.]
 [ 0.  1. 12.  9.]
 [ 0.  0. 20. 13.]]
[01-13 18:30:07][INFO]
[[ 2.  4.  6.  1.]
 [ 0. -1. -7. -8.]
 [ 0.  0.  5.  1.]
 [ 0.  0. 20. 13.]]
[01-13 18:30:07][INFO]
[[ 2.  4.  6.  1.]
 [ 0. -1. -7. -8.]
 [ 0.  0.  5.  1.]
 [ 0.  0.  0.  9.]]
[01-13 18:30:07][INFO]
[[ 2.  4.  6.  1.]
 [ 0. -1. -7. -8.]
 [ 0.  0.  5.  1.]
 [ 0.  0.  0.  9.]]

L de A4:
[[ 1.  0.  0.  0.]
 [ 2.  1.  0.  0.]
 [ 1. -1.  1.  0.]
 [ 3. -0.  4.  1.]]

U de A4:
[[ 2.  4.  6.  1.]
 [ 0. -1. -7. -8.]
 [ 0.  0.  5.  1.]
 [ 0.  0.  0.  9.]]
[01-13 18:30:07][INFO] Sustitución hacia adelante
[01-13 18:30:07][INFO] y =
[[  1.]
 [  0.]
 [  3.]
 [-10.]]
[01-13 18:30:07][INFO] Sustitución hacia atrás
[01-13 18:30:07][INFO] i = 2
```

```
[01-13 18:30:07][INFO] suma = [-1.11111111]
[01-13 18:30:07][INFO] U[i, i] = 5.0
[01-13 18:30:07][INFO] y[i] = [3.]
[01-13 18:30:07][INFO] i = 1
[01-13 18:30:07][INFO] suma = [3.13333333]
[01-13 18:30:07][INFO] U[i, i] = -1.0
[01-13 18:30:07][INFO] y[i] = [0.]
[01-13 18:30:07][INFO] i = 0
[01-13 18:30:07][INFO] suma = [16.35555556]
[01-13 18:30:07][INFO] U[i, i] = 2.0
[01-13 18:30:07][INFO] y[i] = [1.]

Solución para b1 (de A4):
[[1.]
 [2.]
 [4.]
 [5.]]
=> x =
[[-7.67777778]
 [ 3.13333333]
 [ 0.82222222]
 [-1.11111111]]
[01-13 18:30:07][INFO] Sustitución hacia adelante
[01-13 18:30:07][INFO] y =
[[  3.]
 [-11.]
 [-12.]
 [ 45.]]
[01-13 18:30:07][INFO] Sustitución hacia atrás
[01-13 18:30:07][INFO] i = 2
[01-13 18:30:07][INFO] suma = [5.]
[01-13 18:30:07][INFO] U[i, i] = 5.0
[01-13 18:30:07][INFO] y[i] = [-12.]
[01-13 18:30:07][INFO] i = 1
[01-13 18:30:07][INFO] suma = [-16.2]
[01-13 18:30:07][INFO] U[i, i] = -1.0
[01-13 18:30:07][INFO] y[i] = [-11.]
[01-13 18:30:07][INFO] i = 0
[01-13 18:30:07][INFO] suma = [-36.2]
[01-13 18:30:07][INFO] U[i, i] = 2.0
[01-13 18:30:07][INFO] y[i] = [3.]

Solución para b2 (de A4):
[[ 3.]
 [-5.]
 [ 2.]
 [ 6.]]
=> x =
```

```
[[19.6]
 [-5.2]
 [-3.4]
 [ 5. ]]
[01-13 18:30:07][INFO] Sustitución hacia adelante
[01-13 18:30:07][INFO] y =
[[  7.]
 [ -6.]
 [-14.]
 [ 35.]]
[01-13 18:30:07][INFO] Sustitución hacia atrás
[01-13 18:30:07][INFO] i = 2
[01-13 18:30:07][INFO] suma = [3.88888889]
[01-13 18:30:07][INFO] U[i, i] = 5.0
[01-13 18:30:07][INFO] y[i] = [-14.]
[01-13 18:30:07][INFO] i = 1
[01-13 18:30:07][INFO] suma = [-6.06666667]
[01-13 18:30:07][INFO] U[i, i] = -1.0
[01-13 18:30:07][INFO] y[i] = [-6.]
[01-13 18:30:07][INFO] i = 0
[01-13 18:30:07][INFO] suma = [-17.84444444]
[01-13 18:30:07][INFO] U[i, i] = 2.0
[01-13 18:30:07][INFO] y[i] = [7.]

Solución para b3 (de A4):
[[ 7.]
 [ 8.]
 [-1.]
 [ 0.]]
=> x =
[[12.42222222]
 [-0.06666667]
 [-3.57777778]
 [ 3.88888889]]
```